Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Final Examination

Date: December 17, 2025 **Total Points: 50**
Time: 10:30 AM - 01:30 PM **Duration: 180 minutes**

Note: There are eight problems with a total of 64 points. Solve as many as you can. Maximum you can score is 50. Show all your works. Use only the pages provided. The rough works must be done on the pages provided for the same. Do not forget to write your name and roll number. Also circle the location/centre you are attending. Good luck!

Name: ______________________ **Roll No.:** ______________________

Location/Centre: Kolkata Delhi Bangalore

Please DO NOT write any thing below

For Instructor/TAI use only

Problems	1	2	3	4	5	6	7	8	Total
Points									

[Please Only Turn at 11:30 AM]

[Please DO NOT write anything on this page!]

NOTE: The problems are the followings which you will have to solve on the subsequent pages:

1. Suppose three random variables (X, Y, Z) have a *joint trivariate normal* distribution with mean $\mathbf{0} \in \mathbb{R}^3$ and 3×3 *positive definite* matrix Σ as the variance-co-variance matrix, which is given by

$$\Sigma := \begin{pmatrix} 1 & 1/3 & 1/3 \\ 1/3 & 1 & 1/3 \\ 1/3 & 1/3 & 1 \end{pmatrix}$$

   Use the *multidimensional change of variable formula* to find the joint distribution of $X - Y, Y - Z$ and $Z - X$. Your answer should include all details. [8]

2. Suppose X and Y are *i.i.d. standard normal* random variables. Consider the following transformation $(X, Y) \mapsto (R, \Theta)$, where $X = R \sin \Theta$ and $Y = R \cos \Theta$. Find the marginal distributions of R and Θ and the joint distribution of (R, Θ). [$3 + 3 + 2 = 8$]

3. Find the *characteristic function* of $X \sim \text{Poisson}\,(\lambda)$ distribution. Your answer should include all details. [8]

4. Show that if X is a *symmetric* random variable then its *characteristic function* is a real valued continuous function. [$4 + 4 = 8$]

5. Suppose $X_n \stackrel{d}{\longrightarrow} X$ as $n \to \infty$, then show that $\dfrac{X_n}{n} \stackrel{\mathbf{P}}{\to} 0$ as $n \to \infty$. [8]

6. Let $U_1, U_2, \ldots, U_n, \ldots$ be *i.i.d.* Uniform $[0, 1]$. Then show that the sequence of random variables $\left(\prod_{i=1}^{n} U_i\right)^{1/n}$ converges *almost surely*. Find the limit. [$4 + 4 = 8$]

7. Suppose $(A_n)_{n=1}^{\infty}$ is a sequence of events, such that, $\mathbf{P}\,(A_n \text{ i.o. }) > 0$. Then show that there must be two indices $1 \leq i \neq j < \infty$, such that, $A_i \cap A_j \neq \emptyset$. [8]

8. Show that $\mathcal{L}_3 \subseteq \mathcal{L}_2$. Your answer should include all details. [8]

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Rough Work Page

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